

M.A.I.T.R.I.

Mastitis AI-based Tracking & Real-time Inference: An AIoT-Based Early Mastitis Prediction and Decision-Support System for Dairy Cattle

1. Abstract

Bovine mastitis is one of the major health and productivity challenges in dairy farming. It is an inflammation of the mammary gland that can negatively affect animal health, milk production, milk quality, and farm profitability. A major challenge is that mastitis, particularly subclinical mastitis, may develop without obvious visible symptoms and may therefore remain undetected until the condition becomes more severe.

M.A.I.T.R.I. (Mastitis AI-based Tracking & Real-time Inference) is a proposed **Artificial Intelligence of Things (AIoT)** system designed to provide early warning of mastitis risk by continuously monitoring multiple parameters associated with animal health, milk quality, and farm environment.

The system combines a wearable **animal collar**, a **milk monitoring station**, and **environmental monitoring sensors**. The collected data is transmitted through Wi-Fi, LoRa, or GSM to a backend/cloud platform, where it is cleaned, processed, and combined with historical animal records. Machine learning models such as Logistic Regression, Random Forest, XGBoost/LightGBM, and time-series models such as LSTM/GRU can then be evaluated for mastitis-risk prediction.

The intended output is an **individual-animal mastitis risk probability**, potentially providing a **7–14 day early-warning forecast** when sufficient longitudinal training data is available. The result is converted into risk categories such as No Risk, Low Risk, Moderate Risk, and High Risk. Farmers and veterinarians receive alerts through a dashboard/mobile application and can perform appropriate inspection, milk-quality testing, or veterinary intervention.

The system is designed around a **low-cost, modular, scalable architecture** suitable for Indian dairy farms, particularly where expensive centralized livestock-monitoring infrastructure may not be practical.

2. Introduction

Dairy farming is an important component of India's agricultural and rural economy. Maintaining the health of dairy cattle and buffaloes is essential for sustainable milk production and farmer profitability.

Among the diseases affecting dairy animals, **mastitis** is particularly important because it directly affects the mammary gland and therefore milk production and quality.

Mastitis can be caused by different microorganisms entering through the teat canal. Depending on the pathogen and severity, an infected animal may experience:

- Udder inflammation
- Pain and discomfort
- Increased body temperature
- Reduced activity
- Changes in rumination and feeding behaviour
- Changes in milk quality
- Increased Somatic Cell Count (SCC)
- Changes in milk electrical conductivity
- Reduced milk yield

The difficulty is that **subclinical mastitis may not show obvious external symptoms**. Consequently, relying entirely on manual observation can delay detection.

M.A.I.T.R.I. addresses this challenge by shifting from:

Manual observation → disease detection after symptoms

towards:

Continuous monitoring → AI-based risk prediction → early intervention

3. Problem Statement

Mastitis is often detected after visible clinical symptoms appear. By that time, the infection may already have affected milk production and animal health.

Traditional monitoring methods can face several challenges:

1. **Delayed detection**
 - Visible symptoms may appear after the disease has progressed.
2. **Manual monitoring**
 - Farmers cannot continuously observe every animal.
3. **Subclinical mastitis**
 - Animals may appear normal while physiological and milk-quality changes are already occurring.
4. **Economic losses**
 - Reduced milk production and quality can directly affect farmer income.
5. **Treatment expenses**
 - Late intervention can increase veterinary and treatment costs.
6. **Antimicrobial usage**
 - Unnecessary or inappropriate antimicrobial use can contribute to antimicrobial resistance.
7. **Fragmented data**
 - Animal records, milk measurements, environmental conditions, and health history are often stored separately.
8. **Lack of affordable integrated systems**
 - Advanced commercial livestock monitoring systems can be difficult to deploy across smallholder farms.

Core problem

How can real-time animal, milk, environmental, and historical farm data be integrated using AIoT to identify animals at increased risk of mastitis before obvious clinical symptoms appear?

4. Proposed Solution

M.A.I.T.R.I. proposes a **multimodal AIoT monitoring system** consisting of three primary data-collection modules.

4.1 Animal Collar

A wearable collar is used to continuously collect animal-related information.

Possible parameters include:

- Body temperature
- Activity/movement
- Rumination
- Feeding behaviour
- Physiological indicators

The current prototype primarily focuses on:

- ESP32
- Temperature sensing
- MPU6050-based activity monitoring

The final system can incorporate additional sensors depending on cost and validation requirements.

4.2 Milk Monitoring Station

A shared milk-monitoring station can collect milk-related parameters during milking.

Potential parameters include:

- Milk electrical conductivity
- Milk temperature
- Milk pH
- Milk yield/flow
- Somatic Cell Count (SCC)

SCC stands for **Somatic Cell Count** and is an important indicator associated with udder inflammation and mastitis.

Instead of installing expensive SCC hardware on every animal, SCC can initially be incorporated through periodic testing or external/portable testing.

4.3 Farm Environment Monitoring

An environmental node monitors conditions around the cattle shed.

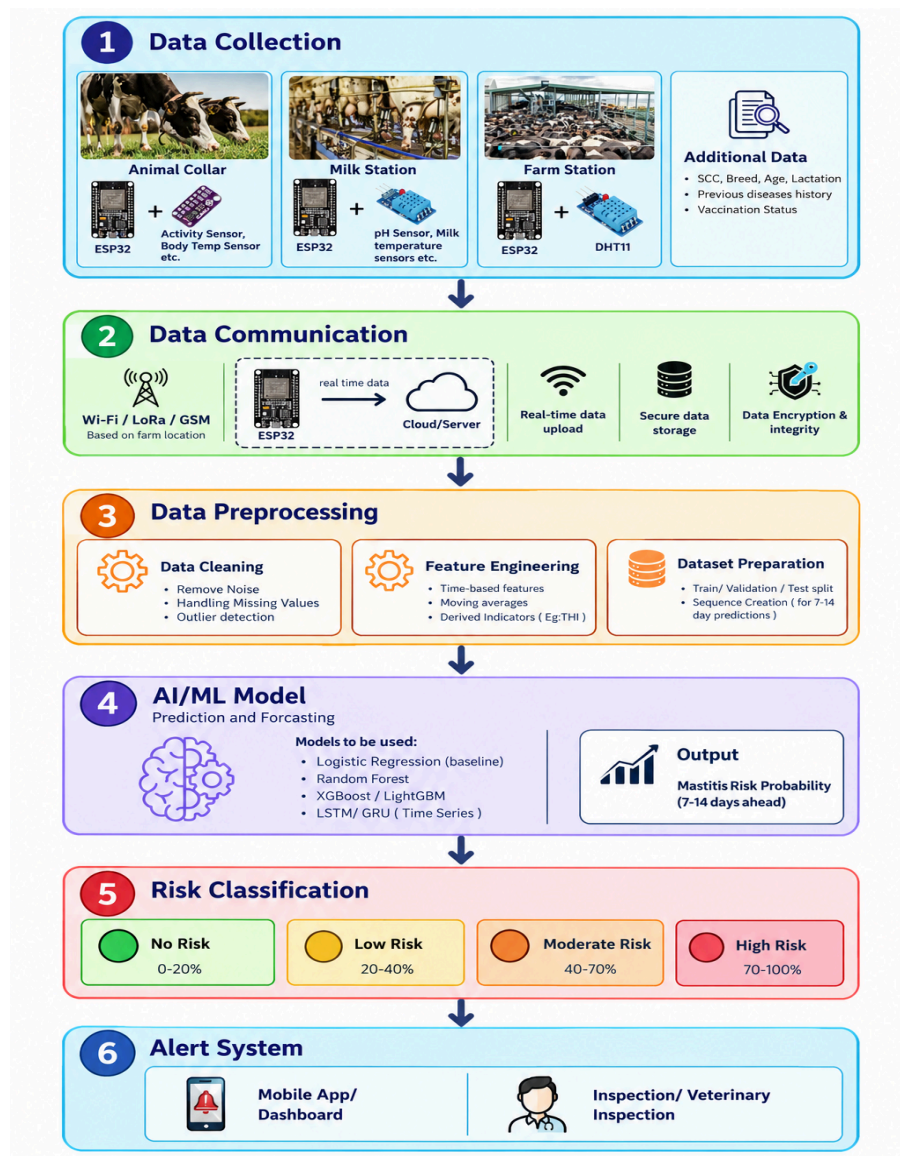
Parameters include:

- Ambient temperature
- Relative humidity
- Temperature-Humidity Index (THI)

THI can be derived from temperature and humidity and can help represent environmental heat-stress conditions.

5. System Architecture

The overall architecture can be represented as:



6. Data Collection

The system uses a **multimodal data strategy**.

Animal data

Parameter	Purpose
Body temperature	Detect physiological abnormalities
Activity	Detect behavioural changes
Rumination	Detect changes in feeding/digestive behaviour
Feeding behaviour	Identify health-related behavioural changes
Historical disease	Individual risk context

Milk Data

Parameter	Purpose
Milk conductivity	Potential indicator of udder/milk changes
Milk temperature	Milk-quality/health context
Milk pH	Milk-quality indicator
Milk yield	Detect production changes
SCC	Important mastitis-associated indicator

Environment Data

Parameter	Purpose
Temperature	Environmental stress
Humidity	Environmental condition
THI	Heat-stress/environmental feature

Historical data

Additional records may include:

- Animal ID
- Breed
- Age
- Lactation number
- Lactation stage
- Previous mastitis history
- Previous diseases
- Vaccination status
- Previous milk yield
- Treatment history
- Farm-management information

7. Data Communication

The system is designed to support different communication methods depending on farm infrastructure.

Wi-Fi

Suitable for:

- Prototype
- Small farms
- Farms with local Wi-Fi

LoRa

Suitable for:

- Larger farms
- Longer-range low-power communication
- Situations where Wi-Fi coverage is limited

GSM

Suitable for:

- Remote farms

- Areas where cellular connectivity is more practical than local Wi-Fi

The ESP32 can collect sensor data and transmit it to a gateway/backend.

8. Data Preprocessing

Raw sensor data cannot be directly given to an ML model.

The system therefore performs preprocessing.

8.1 Data Cleaning

Includes:

- Noise removal
- Missing-value handling
- Duplicate removal
- Outlier detection
- Sensor error detection

8.2 Feature Engineering

Instead of only using raw values, the system can generate:

- Moving averages
- Minimum/maximum values
- Rate of change
- Daily activity trends
- Temperature deviation
- Milk-yield change
- Conductivity trend
- Rumination change
- THI
- Previous mastitis history

For example:

Current temperature

+

Previous 6-hour average

+

Previous 24-hour average

↓

Temperature trend feature

This is especially important because **a single abnormal measurement does not necessarily indicate mastitis.**

9. Dataset Preparation

For genuine 7–14 day prediction, the dataset should ideally be longitudinal.

A simplified structure would be:

Animal ID
Date
Temperature
Activity
Rumination
Milk Yield
Conductivity
SCC
Humidity
THI
Disease History
Mastitis Label

For forecasting:

Day -14
Day -13
Day -12
...
Day -2
Day -1
↓
Prediction
↓
Mastitis in next 7–14 days?

This distinction is important.

A model trained only on:

"Current sensor reading → current mastitis"

does **not** prove 7–14 day forecasting.

The dataset needs enough temporal information to support the forecasting claim.

10. Machine Learning Layer

The proposed system can evaluate several models.

10.1 Logistic Regression

Used as the baseline model.

Advantages:

- Simple
- Fast
- Explainable
- Lightweight
- Suitable for edge deployment

10.2 Random Forest

Suitable for tabular multimodal data.

Advantages:

- Handles nonlinear relationships
- Robust to noisy features
- Relatively easy to interpret
- Can provide feature importance

10.3 XGBoost / LightGBM

Useful for structured/tabular data.

Advantages:

- Strong predictive performance
- Handles nonlinear relationships
- Efficient training

However, deployment directly on an ESP32 may require model optimization.

10.4 LSTM / GRU

Suitable when the data is strongly time-dependent.

Example:

14 days of sensor history



LSTM / GRU



Mastitis probability

The drawback is that these models require more computational resources and sufficiently large longitudinal datasets.

11. TinyML Deployment

One of the proposed future directions is **TinyML-based edge inference**.

Instead of sending every decision to the cloud:

Sensor



ESP32



TinyML Model



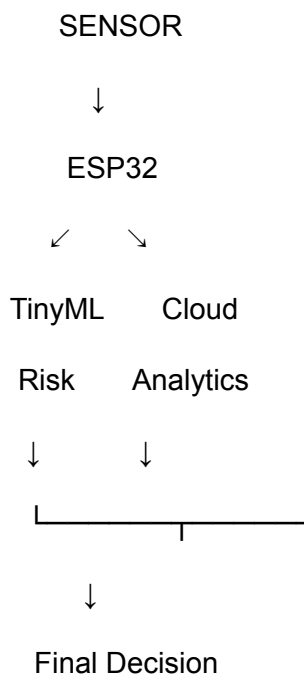
Initial Risk

The system can perform lightweight inference locally.

The cloud can still handle:

- Historical storage
- Advanced analysis
- Model training
- Dashboard
- Model updates

This produces a hybrid architecture:



TinyML should be considered a **deployment optimization**, not a claim that the prototype already has a clinically validated TinyML mastitis model.

12. Risk Classification

The ML model generates a probability/risk score.

A prototype classification can use:

Risk Level	Prototype Score
 No Risk	0–20%
 Low Risk	20–40%
 Moderate Risk	40–70%
 High Risk	70–100%

These thresholds should ultimately be determined through **dataset validation and clinical/farm testing**, rather than assuming that these exact percentages are medically validated.

13. Early Warning System

When an animal crosses a predefined validated risk threshold, the system can generate an alert.

Example:

Animal COW-014 — High Mastitis Risk

Possible recommendation:

"Increased risk detected. Inspect udder and milk quality. Consider SCC/conductivity testing and veterinary evaluation."

The system should **not claim that the sensor has diagnosed mastitis.**

Instead:

AI Prediction



Risk Alert



Milk / SCC Verification



Veterinary Assessment



Confirmed Diagnosis



Treatment / Preventive Action

This is much more scientifically responsible.

14. Dashboard

The M.A.I.T.R.I. dashboard provides a centralized view of herd health.

Dashboard components

- Total animals monitored
- Active sensor devices
- Average temperature
- Activity trends
- Environmental conditions
- Individual animal history
- Risk score
- Risk category
- Sensor telemetry
- Alerts
- Recommendations

Example:

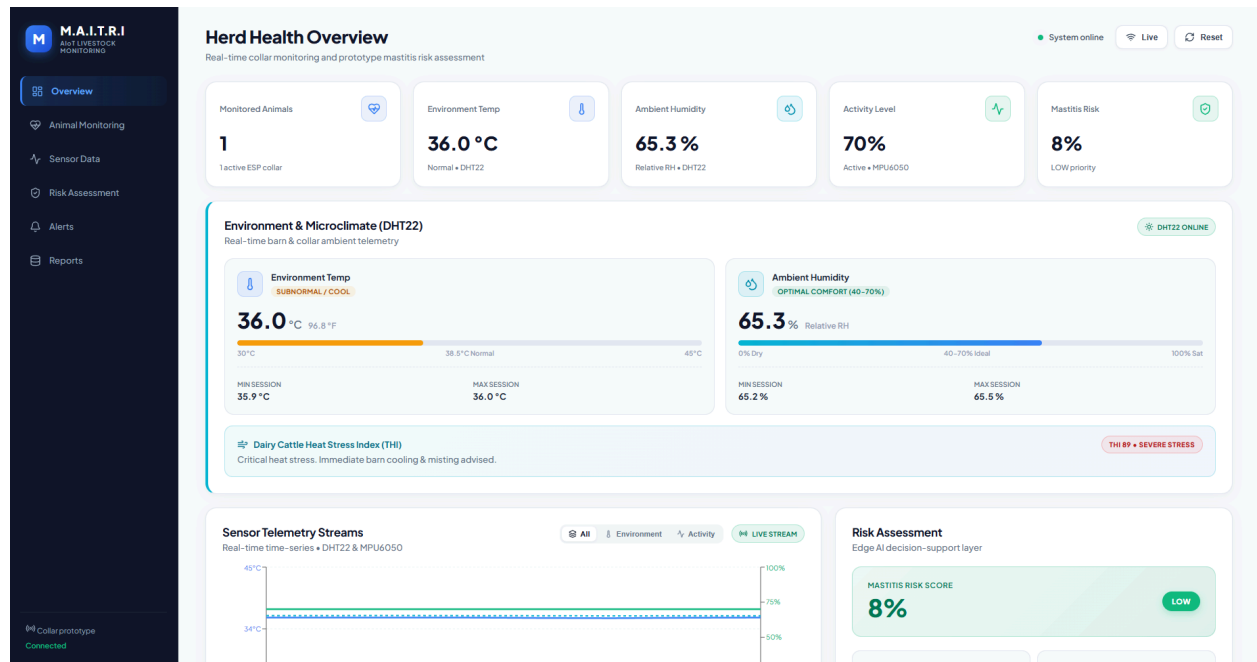


Fig 2 : M.A.I.T.R.I Dashboard User Interface

15. Action and Decision Support

The major purpose of M.A.I.T.R.I. is not merely to produce a graph.

The system should convert prediction into **actionable information**.

Example workflow

AI detects abnormal trend



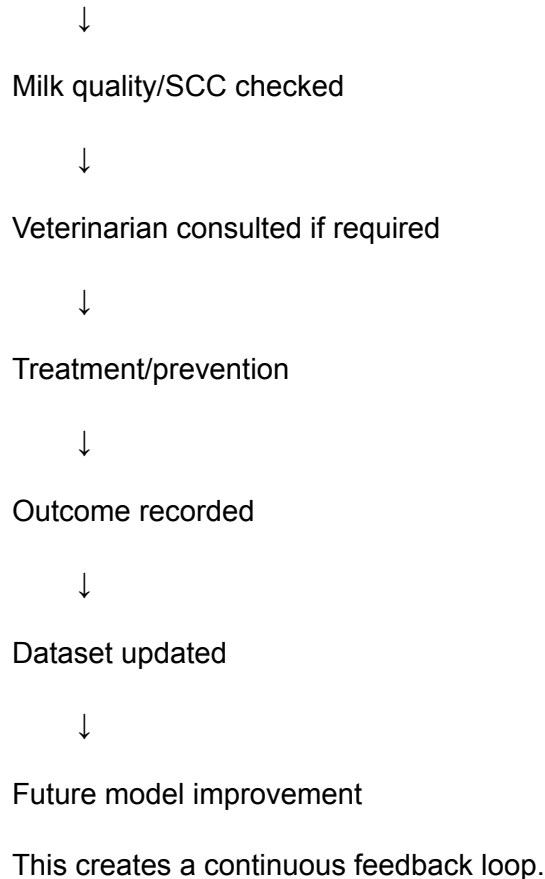
High-risk alert



Farmer receives notification



Animal inspected



16. Innovation and Novelty

The project should **not claim that the smart collar itself is completely novel**, because commercial livestock wearables and research systems already exist.

The stronger novelty lies in the proposed **integration and deployment approach**.

Key differentiating aspects

1. Mastitis-specific prediction

Instead of general livestock monitoring, M.A.I.T.R.I. focuses specifically on mastitis risk.

2. Multimodal data

The system combines:

Animal + Milk + Environment + Historical Data

rather than relying on a single sensor.

3. Early forecasting

The target is early risk prediction rather than simply detecting an already visible clinical condition.

4. Low-cost modular architecture

The system separates:

- Animal wearable
- Shared milk station
- Environment node
- Gateway

This avoids putting every expensive sensor on every animal.

5. TinyML-ready architecture

The model can potentially be optimized for edge inference on resource-constrained hardware.

6. Indian smallholder focus

The architecture is intended to be:

- Affordable
- Modular
- Scalable
- Farmer-friendly
- Adaptable to varying connectivity

17. Comparison with Existing Approaches

The table below compares our prototype with the Traditional Monitoring approach as well as the Commercial Smart System that are available in India,

Aspect	Traditional Monitoring	Commercial Smart Systems	M.A.I.T.R.I.
Continuous monitoring	Limited	Yes	Yes
Animal wearable	No/limited	Yes	Yes
Milk monitoring	Manual/separate	Available	Integrated
Environmental data	Usually separate	Varies	Integrated

Historical records	Manual	Varies	Integrated
Mastitis-specific AI	Limited	Some systems	Core objective
Multimodal approach	Limited	Varies	Core architecture
Early forecasting	Limited	Varies	Targeted
TinyML	Usually absent	Varies	Planned
Low-cost modular design	—	Varies	Core design objective
Farmer dashboard	Limited	Yes	Yes
Veterinary decision support	Manual	Varies	Integrated

18. Technological Stacks

Hardware Technological Stacks

Animal Collar

- ESP32
- DS18B20 or appropriate body-temperature sensor
- MPU6050
- Optional rumination sensor
- Battery
- Charging/protection circuit
- Custom PCB
- Waterproof enclosure
- Adjustable collar

Milk Station

- ESP32
- Conductivity sensor
- pH sensor
- Milk-temperature sensor
- Yield/flow measurement
- Optional SCC measurement interface

Environment Station

- ESP32
- DHT22/SHT31/BME280
- Temperature
- Humidity

Communication

- Wi-Fi
- LoRa
- GSM

depending on deployment requirements.

Software Technology Stack

A practical stack for your project is:

Layer	Technology
Microcontroller	ESP32
Firmware	Arduino IDE / C++
Data communication	Wi-Fi / LoRa / GSM
Backend	Node.js + Express
API	REST API
Database	MongoDB / Firebase

ML development	Python
ML	Scikit-learn / XGBoost / TensorFlow
Deep learning	TensorFlow
Edge AI	TensorFlow Lite Micro / TinyML
Frontend	React
Visualization	Recharts / similar charting library
Version control	Git/GitHub
Development	VS Code
Deployment	Cloud server / Docker where required

For our current prototype, we havent implemented few things till now. This is our proposed architecture.

19. Prototype Implementation

The current prototype focuses on proving the AIoT pipeline.

Current hardware

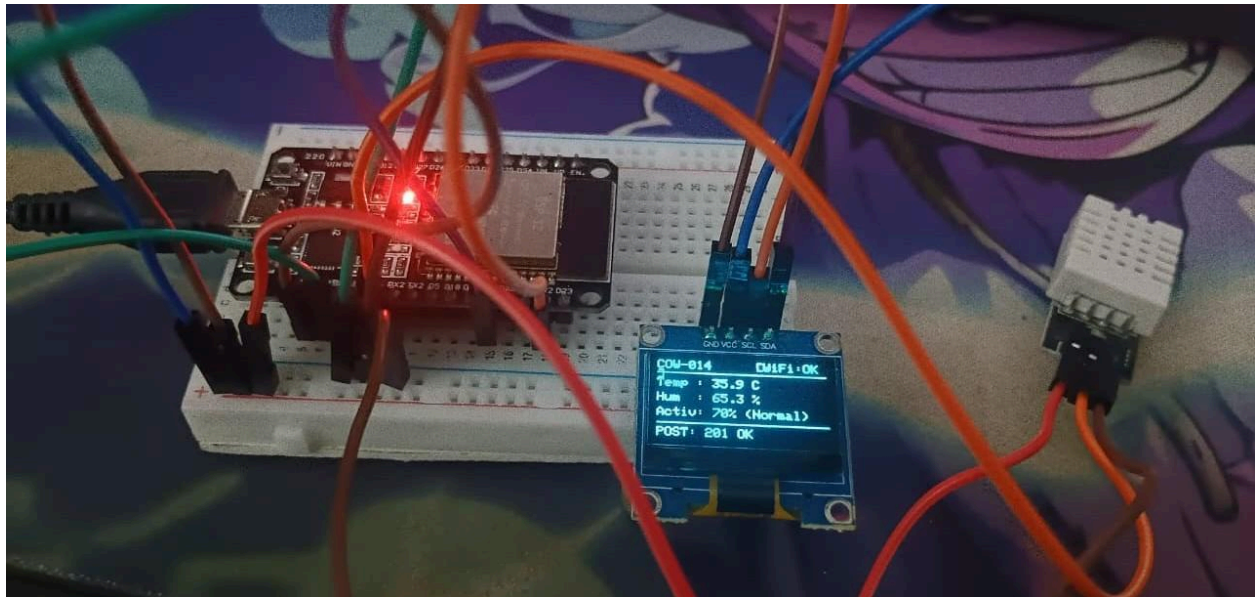


Fig 3: Environment Module Prototype

The DHT22 is used for **environmental temperature/humidity**, not animal body temperature.

The final animal collar can use a dedicated body-temperature sensor such as a waterproof DS18B20 or another suitable validated sensor.

Current software

The dashboard can display:

- Sensor values
- Trends
- Animal ID
- Risk indicator
- Monitoring status

For the current demonstration, simulated/recorded data can be used where genuine longitudinal mastitis data is unavailable.

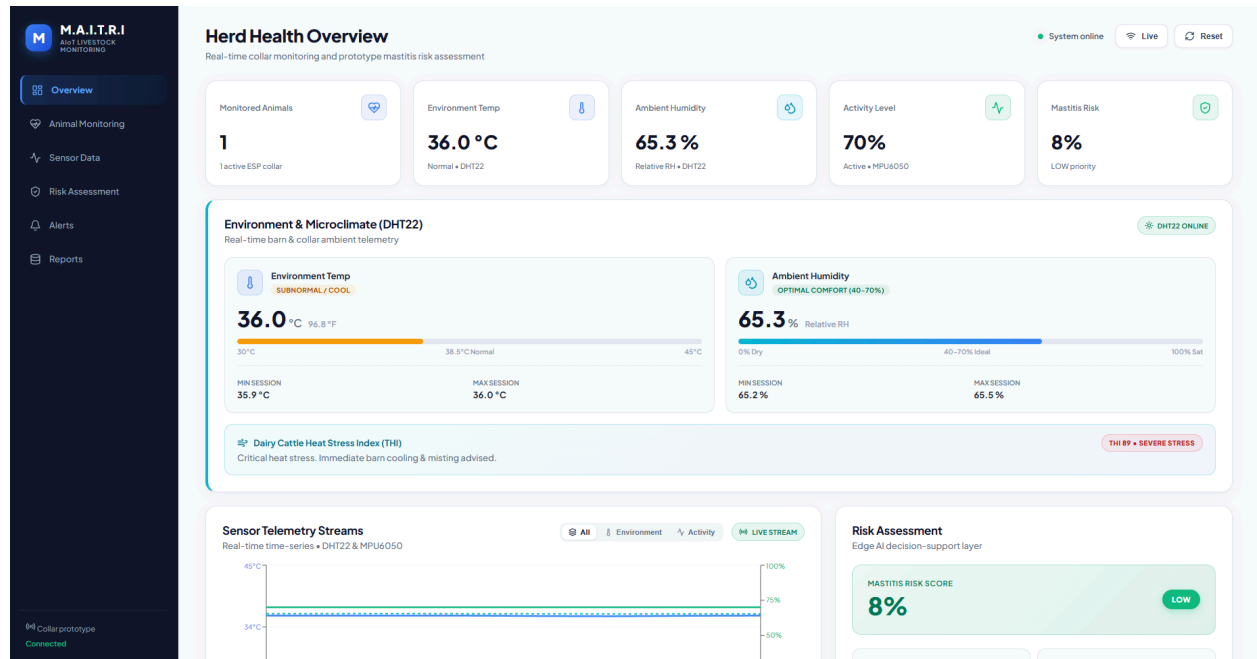


Fig 4: Dashboard User Interface

20. Cost Analysis

A practical target BOM is:

Module	Estimated Cost
Smart animal collar	₹2,000–₹3,500
Milk monitoring station	₹5,000–₹10,000
Environment node	₹1,000–₹2,000
Gateway/communication	₹1,500–₹4,000

Power & enclosure	₹500–₹1,000
Software/AI prototype	Low incremental cost
Small-farm complete setup	~₹11,000–₹25,500

The architecture is intentionally modular.

For example, a farm with 20 animals does **not** need 20 milk stations.

It can have:

20 × Animal Collar

+

1 × Milk Station

+

1–2 × Environment Nodes

+

1 × Gateway

This reduces the effective per-animal infrastructure cost.

21. Economic Benefits

The expected economic benefits include:

Reduced milk loss

Earlier intervention can help reduce the duration/severity of mastitis-related production losses.

Reduced treatment cost

Earlier identification can support timely veterinary intervention.

Reduced unnecessary antimicrobial use

Better risk assessment may help avoid indiscriminate treatment, subject to veterinary diagnosis.

Improved herd productivity

Continuous monitoring can identify animals requiring attention.

Better resource allocation

Farmers and veterinarians can prioritize high-risk animals rather than manually checking every animal with equal intensity.

22. Feasibility

Technical Feasibility

ESP32-based sensor systems are inexpensive and widely available.

Challenge: sensor accuracy and environmental durability.

Mitigation:

- Sensor calibration
- Filtering
- Waterproof enclosures
- Periodic maintenance
- Sensor validation

AI/ML Feasibility

Machine-learning models can process structured animal, milk, environmental, and historical data.

Challenge: obtaining a sufficiently large longitudinal mastitis dataset.

Mitigation:

- Collaborate with dairy farms/veterinary institutions
- Collect longitudinal data
- Use multiple models
- Validate using unseen animals/farms
- Avoid claiming 7–14 day forecasting until validated

Connectivity Feasibility

Challenge: rural farms may have unreliable internet.

Mitigation:

- LoRa
- GSM
- Local storage
- Store-and-forward architecture

Economic Feasibility

Using shared milk/environment infrastructure rather than duplicating expensive equipment for every cow helps reduce cost.

Farmer Adoption

The interface should remain simple:

 Cow 021 — Normal

 Cow 014 — Monitor

 Cow 018 — Inspect

 Cow 007 — Veterinary Attention

The farmer does not need to understand the underlying ML algorithm.

23. Risks and Limitations

The project has several technical limitations.

1. Dataset limitation

The biggest current challenge is obtaining a sufficiently large, high-quality longitudinal dataset.

2. Sensor variability

Sensors may behave differently across:

- breeds
- farms
- weather conditions
- sensor placements

3. False positives

A cow may show:

- increased temperature
- reduced activity
- reduced rumination

because of diseases other than mastitis or because of heat stress.

Therefore:

Sensor abnormalities should not be interpreted as definitive mastitis diagnosis.

4. False negatives

A model may fail to identify some cases.

Therefore, veterinary confirmation remains necessary.

5. Forecasting uncertainty

A 7–14 day prediction is significantly more challenging than current-condition classification.

It requires appropriate longitudinal data and rigorous validation.

24. Future Scope

The project can be expanded in several directions.

Phase 1 — Prototype

- ESP32
- Temperature
- Activity
- Environmental sensing
- OLED
- Wi-Fi
- Dashboard

Phase 2 — Multimodal Monitoring

Add:

- Milk conductivity
- Milk temperature
- pH
- Milk yield
- SCC data
- Historical records

Phase 3 — Machine Learning

Train and compare:

- Logistic Regression
- Random Forest
- XGBoost/LightGBM
- LSTM/GRU

Evaluate:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion matrix

For medical/animal-health prediction, **recall and false-negative rate are particularly important**, because missing a genuinely high-risk animal can be more consequential than generating an extra alert.

Phase 4 — TinyML

Optimize the selected model for:

- ESP32
- Memory
- Flash
- Inference time
- Energy consumption

Phase 5 — Field Validation

Test across:

- Multiple farms
- Different breeds
- Different climates
- Different herd sizes

Phase 6 — Commercial Deployment

Develop:

- Rugged PCB
- Waterproof collar
- Long-life battery
- LoRa/GSM communication
- Farmer mobile application
- Cloud platform
- Veterinary integration

25. Expected Outcomes

The expected outcomes of M.A.I.T.R.I. are:

Animal Health

- Earlier identification of potentially affected animals
- Improved monitoring
- Timely veterinary intervention

Milk Production

- Reduced production losses
- Improved milk-quality management

Farmer Economics

- Lower avoidable treatment costs
- Reduced production losses
- Better herd-management decisions

Public Health

- Better antimicrobial stewardship
- Improved milk-quality monitoring
- Contribution toward reducing antimicrobial resistance risks

Technology

- Affordable AIoT livestock monitoring
- Data-driven precision dairy farming
- Edge-AI/TinyML deployment potential

26. Project Workflow — Complete

The complete system can be summarized as:

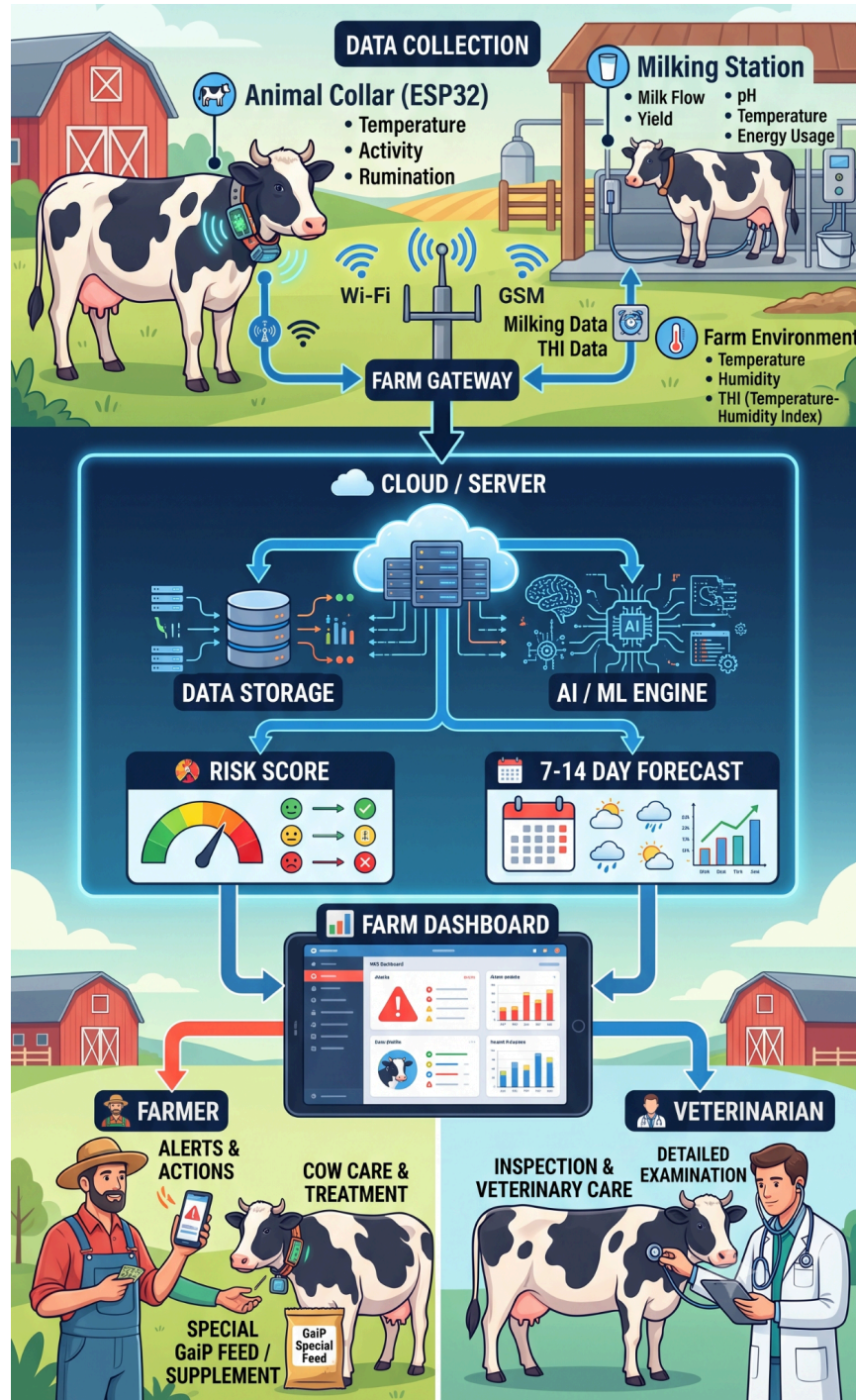


Fig 5: Workflow Diagram

27. Conclusion

M.A.I.T.R.I. proposes an **AIoT-based approach to proactive dairy-animal health management**, with bovine mastitis as the primary target condition.

Rather than relying exclusively on visible clinical symptoms or manual observation, the proposed system combines **animal health indicators, behavioural information, milk-quality parameters, environmental conditions, and historical records**.

The architecture consists of a wearable animal collar, shared milk-monitoring infrastructure, environmental monitoring nodes, wireless communication, cloud/backend processing, machine-learning models, and a farmer/veterinarian decision-support interface.

The major strength of the project is not simply the use of an IoT collar. Its intended differentiation lies in the **integration of multimodal data for mastitis-specific risk forecasting**, combined with a **low-cost, modular architecture and future TinyML-based edge deployment**.